

EXPERIMENT NO : 12

Experiment to detect outliers in a given data set.

PROGRAM:-

```
import numpy as np
```

```
array=np.random.randint(1,100,16)
```

```
array
```

OUTPUT:-

```
array([60, 11, 99, 54, 90, 60, 16, 41, 43, 39, 84, 13, 18, 34, 43, 70])
```

```
float(array.mean())
```

OUTPUT:-

```
48.4375
```

```
float(np.percentile(array,25))
```

OUTPUT:-

```
30.0
```

```
float(np.percentile(array,50))
```

OUTPUT:-

```
43.0
```

```
float(np.percentile(array,75))
```

OUTPUT:-

```
62.5
```

```
float(np.percentile(array,100))
```

OUTPUT:-

99.0

```
def outDetection(array):
```

```
    sorted(array)
```

```
    Q1,Q3=np.percentile(array,[25,75])
```

```
    IQR=Q3-Q1
```

```
    lr=float(Q1-(1.5*IQR))
```

```
    ur=float(Q3+(1.5*IQR))
```

```
    return lr,ur
```

```
lr,ur=outDetection(array)
```

```
lr,ur
```

OUTPUT:-

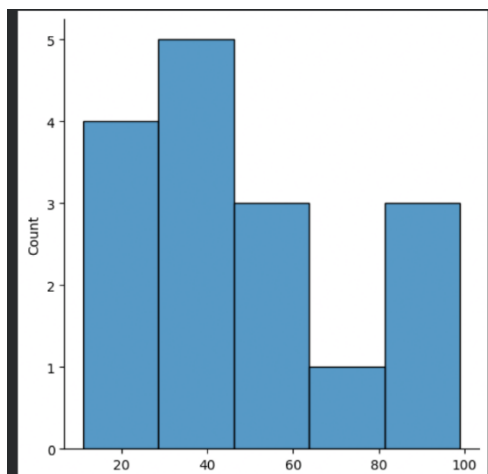
(-18.75, 111.25)

```
import seaborn as sns
```

```
%matplotlib inline
```

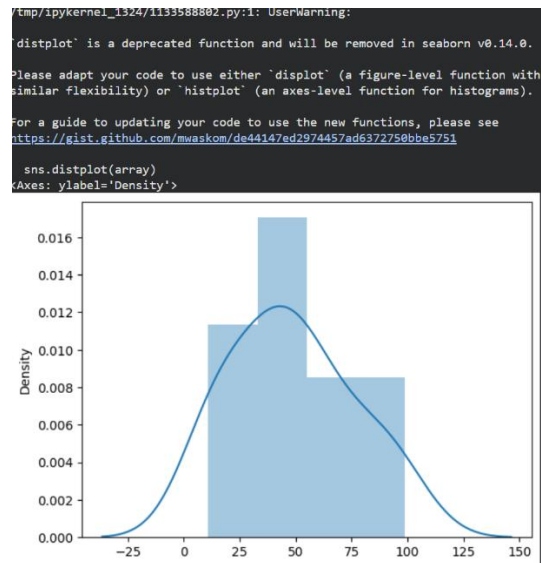
```
sns.displot(array)
```

OUTPUT:-



```
sns.distplot(array)
```

OUTPUT:-



```
new_array=array[(array>lr) & (array<ur)]
```

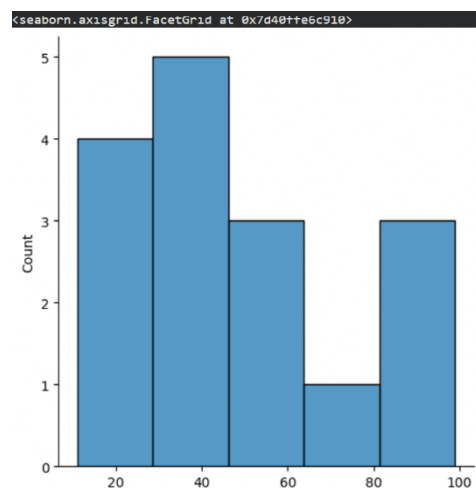
```
new_array
```

OUTPUT:-

```
array([60, 11, 99, 54, 90, 60, 16, 41, 43, 39, 84, 13, 18, 34, 43, 70])
```

```
sns.displot(new_array)
```

OUTPUT:-



```
lr1,ur1=outDetection(new_array)
```

```
lr1,ur1
```

OUTPUT:-

```
(-18.75, 111.25)
```

```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
```

```
final_array
```

OUTPUT:-

```
array([60, 11, 99, 54, 90, 60, 16, 41, 43, 39, 84, 13, 18, 34, 43, 70])
```

```
sns.distplot(final_array)
```

OUTPUT:-

